

Module Description

MW2468: Logistics Engineering in Production Systems and Supply Chain Management

Chair of Chair of Materials Handling, Material Flow, Logistics
(Prof. Fottner)

Module level: Bachelor	Language: English	Module duration: one semester	Occurrence: winter semester
Credits*: 5	Total number of hours: 150	Self-study hours: 45	Contact hours: 105

* The number of credits can vary depending on the corresponding SPO version. The valid number is always indicated on the Transcript of Records or the Performance Record.

Description of achievement and assessment methods:

Students apply the lecture's contents in a written exam (duration: 90 minutes) with questions and calculation tasks. The only aid allowed is a non-programmable calculator. In this way, students demonstrate different abilities: to analyze logistics systems, logistics processes and logistics structures; to apply methods for planning of such structures; to understand the key functions of physical logistics.

The module cannot be taken in combination with "Materialfluss und Logistik" [MW1921]!

Possibility of re-taking:

In the next semester: Yes

At the end of the semester: No

(Recommended) requirements:

no

Contents:

From a higher point of view, the module explains the main principles and goals of logistics engineering, as well as key indicators and impact factors of logistics. Technical processes are explained for a better understanding of production systems, distribution centers and material supply in production systems. Common structures of production and distribution are presented along with according control strategies and technologies. Besides key functions of material flow-transportation, distribution/consolidation, storage, order picking and handling-methods to model material flow systems are taught, e. g. flow charts, graphs, material flow matrices and layouts. Methods to analyze system behavior complete the module; they comprise static dimensioning, event-discrete simulation, emulation, queuing theory and the concept of availability and reliability of technical systems. Furthermore, we give an overview on how data can be available on different levels of logistics systems to enable smart factories.

Technologies for the operations in a smart factory are discussed, comprising control and design strategies (flow shop vs. job shop, modular factories, decentralized and autonomous controls), localization, identification and mobile robots.

Additionally, the module contains the following contents:

- Logistics systems: Design guidelines; logistical processes, functions, and structures; logistical networks; methods for planning logistical structures
- Logistics management: Control and coordination in logistics systems, information management

Study goals:

Having completed the module, students know about key tasks and aims of logistics. They are able to analyze logistics systems, logistical processes and logistical structures. Furthermore, they can apply methods to plan logistical structures and know means of control and coordination in logistics systems and concepts of information management. They know a variety of technologies for smart factories along with their benefits and boundaries.

In addition, students understand the key functions of physical logistics and are able to apply methods to depict material flow and to dimension and evaluate logistics systems.

Teaching and learning methods:

Contents are explained by lectures and by exemplary applications from industrial practice. Supporting the lectures, students have access to a detailed collection of slides, exercises and sample solutions.

In tutorials, exercises demonstrate the applicability of the lectures' theoretical contents.

All documents and further information are accessible online and free via elearning. During office hours of scientific staff, individual questions and problems can be discussed.

Media formats:

Lectures: Talk with tablet and projector, board and overhead projector; printed scriptum (fee-based)

Online documents: Documents for exercises with sample solutions; scriptum (digital as PDF, free of charge)

Literature:

Literature:

Aggteleky, B.: Fabrikplanung: Werksentwicklung und Betriebsrationalisierung, Band 1-3. München, Wien: Hanser 1987 (Band 1) und 1990 (Band 2 und 3)

Arnold, D.: Materialflusslehre. Braunschweig, Wiesbaden: Vieweg, 1998

Dangelmaier, W.: Fertigungsplanung. Düsseldorf: VDI-Verlag, 2001

Gudehus, T.: Logistik: Grundlagen, Strategien, Anwendungen. Berlin u.a.: Springer, 2005

Großeschallau, W.: Materialflussrechnung. Berlin u.a.: Springer, 1984

Kettner, H., Schmidt, J., Greim, H.-R.: Leitfaden der systematischen Fabrikplanung. München, Wien: Hanser, 1984

Jünemann, R.: Materialfluss und Logistik: Systemtechnische Grundlagen mit Praxisbeispielen. Berlin u.a.: Springer, 1998

Jünemann, R., Schmidt, T.: Materialflusssysteme: Systemtechnische Grundlagen. Berlin u.a.: Springer, 1999

Pfohl, H.-C.: Logistiksysteme: Betriebswirtschaftliche Grundlagen. Berlin u.a.: Springer, 2004

VDI-Gesellschaft Fördertechnik Materialfluss Logistik (Hrsg.). VDI-Handbuch Materialfluss und Fördertechnik: Band 1 8. Düsseldorf: VDI-Verlag

Wildemann, H.: Logistik Prozessmanagement. München: TCW Transfer-Centrum, 2005

Wiendahl, H.-P.: Fertigungsregelung: Logistische Beherrschung von Fertigungsabläufen auf Basis des Trichtermodells. München, Wien: Hanser, 1997

Responsible for the module:

Fottner, Johannes; Prof. Dr.-Ing.: j.fottner@tum.de

Courses (Type, SH) Lecturer:

0000002543 Logistics Engineering in Production Systems and Supply Chain Management (2SWS VO, WS 2024/25)
Aleo O [L], Fottner J (Aleo O, Wang Z)

0000002759 Logistics Engineering in Production Systems and Supply Chain Management (2SWS UE, WS 2024/25)
Aleo O [L], Fottner J (Aleo O, Wang Z), Gao L

For further information about this module and its allocation to the curriculum see:
<https://campus.tum.de/tumonline/wbModHb.wbShowMHBReadOnly?pKnotenNr=2657178>

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